Light

# GCSE Computer Science — Algorithms: Searching and Sorting (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A programmer uses linear search to find the number 7 in the list [7, 3, 9, 2].
Which position is checked first?

- A. Position 0
- B. Position 1
- C. Position 2
- D. Position 3

Q2 (1 mark)

A student attempts to use binary search on the list [5, 2, 8, 1, 9].
Why might this search produce incorrect results?

- ○ **A.** The list contains too few items
- ○ **B.** The list contains odd numbers
- ○ **C.** Binary search only works on numbers greater than 10
- ○ **D.** The list is not sorted

Q3 (1 mark)

In bubble sort, larger values move toward which end of the list during sorting into ascending order?

- ○ **A.** The left end (start)
- ○ **B.** The middle
- ○ **C.** The right end (end)
- ○ **D.** They stay in their original positions

Q4 (1 mark)

In merge sort's 'divide and conquer' approach, what does 'conquer' refer to?

- ○ **A.** Deleting unwanted elements
- ○ **B.** Counting the number of items
- ○ **C.** Reversing the list order
- ○ **D.** Merging sublists back together in sorted order

Q5 (1 mark)

What happens when linear search is performed on an empty list?

- ○ **A.** The search returns that the item was not found
- ○ **B.** The algorithm crashes
- ○ **C.** The search returns the first position
- ○ **D.** The algorithm runs forever

Q6 (1 mark)

A binary search examines a list with indices 0 to 9 (10 items). Using integer division, what is the index of the middle element checked first?

- ○ **A.** 5
- ○ **B.** 4
- ○ **C.** 0
- ○ **D.** 9

Q7 (1 mark)

During bubble sort on a list of 8 items, which pairs of positions are compared during the first pass?

- ○ **A.** Only positions 0 and 7
- ○ **B.** All possible pairs of positions
- ○ **C.** Only positions 0 and 4

- ○ **D.** Positions 0 and 1, then 1 and 2, then 2 and 3, continuing to 6 and 7

Q8 (1 mark)

A list of 16 items is being divided using merge sort. How many times must the list be divided before reaching single-element sublists?

- ○ **A.** 2 times
- ○ **B.** 8 times
- ○ **C.** 16 times
- ○ **D.** 4 times

Q9 (1 mark)

The list [3, 7, 3, 9, 3] is searched using linear search to find the value 3. Which occurrence of 3 will be found first?

- ○ **A.** The one at position 4
- ○ **B.** The one at position 2
- ○ **C.** The one at position 0
- ○ **D.** All occurrences simultaneously

Q10 (1 mark)

Binary search is used to find the name 'Emma' in an alphabetically sorted list of student names.
Why can binary search work on text data?

- ○ **A.** Because text data is always shorter than numerical data
- ○ **B.** Because binary search converts text to numbers first
- ○ **C.** Because text data is stored in binary format
- ○ **D.** Because text characters can be compared alphabetically, establishing order

Q11 (1 mark)

A list is already sorted except for one element that is slightly out of place. How does bubble sort perform on this nearly-sorted list?

- ○ **A.** Bubble sort cannot handle nearly-sorted lists
- ○ **B.** Bubble sort performs worst on nearly-sorted lists
- ○ **C.** Bubble sort performs well, requiring fewer passes than worst case
- ○ **D.** Bubble sort treats it the same as a fully reversed list

Q12 (1 mark)

Two students have the same test score. After sorting by score using merge sort, their relative order from the original list is preserved. What property does this demonstrate?

- ○ **A.** Efficiency
- ○ **B.** Recursion
- ○ **C.** Stability

- ○ **D.** Complexity

Q13 (2 marks)

A list contains 50 items. The target value is at position 30.
How many comparisons does linear search need to find it?

- ○ **A.** 30
- ○ **B.** 31
- ○ **C.** 50
- ○ **D.** 25

Q14 (2 marks)

Binary search examines the sorted list [10, 20, 30, 40, 50, 60, 70] looking for 20. After comparing with the middle element (40), which half is searched next?

- ○ **A.** The left half containing [10, 20, 30]
- ○ **B.** The right half containing [50, 60, 70]
- ○ **C.** Both halves simultaneously
- ○ **D.** The middle element is searched again

Q15 (2 marks)

The list [2, 5, 1, 3] is sorted using bubble sort.
After the first complete pass, what is the state of the list?

- ○ **A.** [1, 2, 3, 5]
- ○ **B.** [2, 1, 3, 5]
- ○ **C.** [1, 2, 5, 3]
- ○ **D.** [5, 3, 2, 1]

Q16 (2 marks)

Two sorted sublists [2, 7, 9] and [3, 4] are being merged.
What are the first three elements added to the merged result?

- ○ **A.** 2, 3, 4
- ○ **B.** 2, 7, 9
- ○ **C.** 3, 4, 2
- ○ **D.** 2, 3, 7

Q17 (2 marks)

Binary search looks for the value 15 in the sorted list [10, 20, 30, 40].
The search compares 15 with 20, then with 10.
What does the algorithm conclude?

- ○ **A.** The value 15 is at position 1
- ○ **B.** The value 15 is at position 0
- ○ **C.** The search should continue in the right half
- ○ **D.** The value 15 is not in the list

Q18 (2 marks)

The list [4, 3, 2, 1] is sorted using bubble sort into ascending order.
What is the total number of swaps needed across all passes?

- ○ **A.** 3
- ○ **B.** 4
- ○ **C.** 6
- ○ **D.** 10

Q19 (2 marks)

A list of 8 items is fully divided using merge sort.
How many single-element sublists exist after the divide phase is complete?

- ○ **A.** 3
- ○ **B.** 4
- ○ **C.** 8
- ○ **D.** 16

Q20 (2 marks)

A programmer needs to sort a list of only 5 items. A colleague suggests using merge sort because it is efficient.
What is a reasonable response?

- ○ **A.** Merge sort cannot sort lists with fewer than 10 items
- ○ **B.** Merge sort is always the best choice regardless of list size

- ○ **C.** Five items is too few to require any sorting algorithm
- ○ **D.** For 5 items, simpler algorithms like bubble sort work fine and the efficiency difference is negligible

Q21 (2 marks)

A sorted list contains 500 items. Binary search finds the target after exactly 9 comparisons.
What does this tell us about the target's position?

- ○ **A.** The target was definitely not in the list
- ○ **B.** The target was at the first position
- ○ **C.** The target was at a position requiring the algorithm to narrow down through 9 halvings
- ○ **D.** The target was at the middle position

Q22 (2 marks)

An optimised bubble sort with early termination is run on a list that is already sorted.
How many complete passes are needed?

- ○ **A.** Zero passes — the algorithm detects sorting without any comparisons
- ○ **B.** One pass to verify the list is sorted
- ○ **C.** The same number of passes as worst case
- ○ **D.** Half as many passes as the list has elements

Q23 (2 marks)

A programmer is sorting a very large list on a device with limited memory. They are concerned about merge sort's memory usage.
Why is this concern valid?

- ○ **A.** Merge sort deletes original data, requiring backup copies
- ○ **B.** Merge sort needs temporary storage for sublists and merged results, proportional to list size
- ○ **C.** Merge sort stores all comparisons in memory
- ○ **D.** Merge sort requires memory for recursive function calls only, which is minimal

Q24 (3 marks)

A teacher claims that linear search is sometimes preferable to binary search.
In which scenario is this claim valid?

- ○ **A.** When searching a list of 1 million sorted items repeatedly
- ○ **B.** When the data is unsorted and will only be searched once
- ○ **C.** When the target is known to be in the middle of a sorted list
- ○ **D.** When memory is very limited and cannot store sorted data

Q25 (3 marks)

A binary search is performed on [1, 3, 5, 7, 9, 11, 13, 15] to find the value 1.
Which sequence of comparisons occurs?

- ○ **A.** 7, 3, 1

- **B.** 7, 5, 3, 1
- **C.** 1
- **D.** 8, 4, 2, 1

Q26 (3 marks)

The list [3, 1, 4, 2] is sorted using bubble sort.
What is the total number of comparisons made across all passes until the list is sorted?

- **A.** 6
- **B.** 9
- **C.** 12
- **D.** 3

Q27 (3 marks)

The list [6, 2, 5, 3] is sorted using merge sort.
What is the sequence of merge operations?

- **A.** Merge [6][2]→[2,6], merge [5][3]→[3,5], merge [2,6] [3,5]→[2,3,5,6]
- **B.** Merge [6,2][5,3]→[2,3,5,6]
- **C.** Merge [6][2][5][3]→[2,3,5,6]
- **D.** Merge [2,6][3,5]→[2,3,5,6] only

Q28 (3 marks)

A database stores 100,000 customer records sorted by surname. The system needs to find specific customers hundreds of times per second. Which search algorithm is most appropriate and why?

- ○ **A.** Linear search, because it handles large lists better
- ○ **B.** Bubble sort followed by linear search for each query
- ○ **C.** Binary search, because logarithmic time allows hundreds of searches per second on sorted data
- ○ **D.** Linear search, because binary search cannot handle text data like surnames

Q29 (3 marks)

A programmer analyses bubble sort's worst-case performance on a list of n items.
Which statement correctly describes the worst case?

- ○ **A.** The worst case requires n comparisons total
- ○ **B.** The worst case occurs when the list is already sorted
- ○ **C.** The worst case requires approximately $n^2 / 2$ comparisons, occurring when the list is in reverse order
- ○ **D.** The worst case cannot be determined without knowing the specific values

Q30 (3 marks)

A student asks why merge sort is often preferred over bubble sort for large datasets despite bubble sort being simpler.
Which explanation correctly addresses this question?

- ○ **A.** Merge sort uses less memory than bubble sort, making it more practical
- ○ **B.** Merge sort's O(n log n) time complexity grows much more slowly than bubble sort's $O(n^2)$, making it dramatically faster for large datasets despite higher overhead
- ○ **C.** Bubble sort cannot sort lists with more than 1000 items
- ○ **D.** Merge sort was invented more recently and is therefore superior

Total: **55** marks • Time: **30 min**

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